

Adaptive Parameter Synthesis Agent Architecture

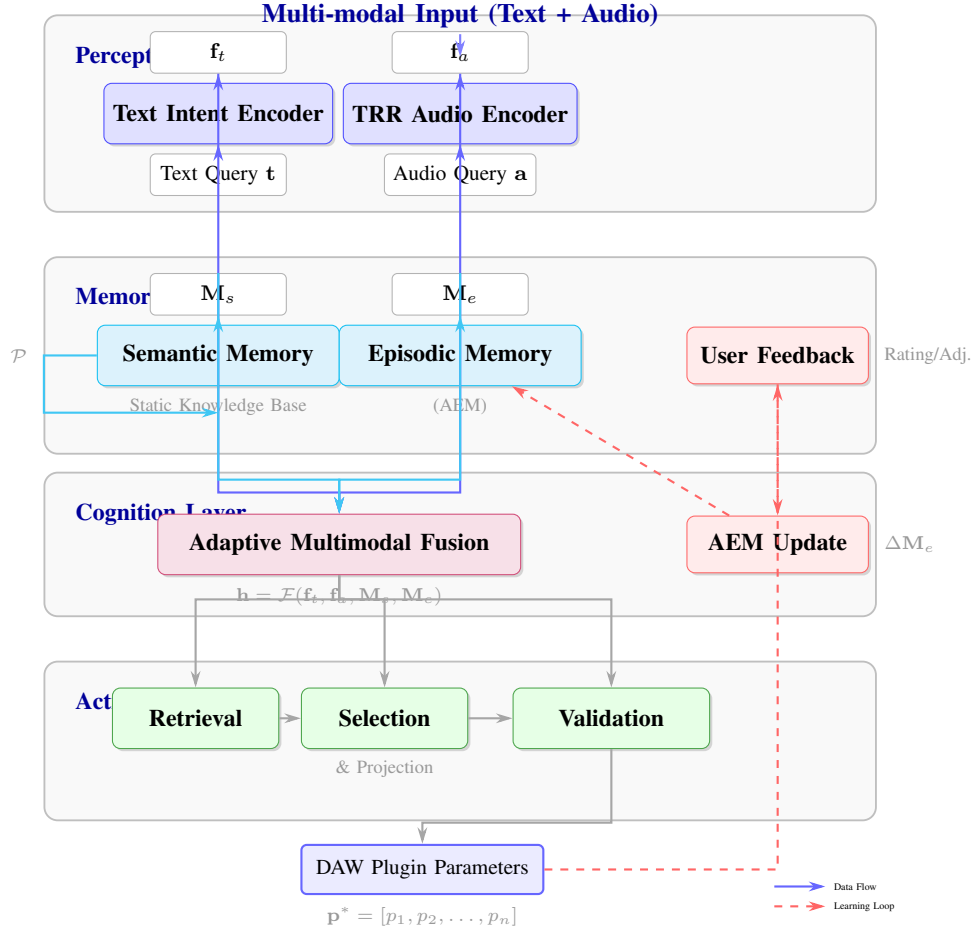


Fig. 1. Architecture of the proposed Adaptive Parameter Synthesis Agent. The system processes multi-modal inputs (text and audio) through a perception layer, retrieves relevant knowledge from semantic and episodic memory, fuses information via adaptive multimodal fusion, and executes parameter synthesis through retrieval, selection, and validation. A learning loop enables continuous improvement through user feedback and AEM updates.